

BERNARD BOAKYE APPIAH

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Aspiring Critical Infrastructure & OT Cybersecurity Professional

Permanent Resident of Canada | Authorized to work in Canada

PROFESSIONAL SUMMARY

Cybersecurity Operations and Management graduate student and Telecommunications Engineering graduate with hands-on experience in network security labs, risk assessment, firewall segmentation, and infrastructure-focused projects. Skilled in packet analysis, network segmentation, NIST CSF and MITRE ATT&CK-based risk assessment, Linux systems, and clear technical documentation. Motivated by real-world breaches and an engineering background to protect operational technology, critical infrastructure, and industrial control environments. Seeking cybersecurity intern, SOC analyst intern, network security, GRC, OT/security operations, and IT security analyst opportunities.

KEY STRENGTHS

- Risk assessment using NIST CSF and MITRE ATT&CK threat mapping
- Network segmentation, firewall rules, and network device configuration
- Packet analysis and traffic validation with Wireshark, Tcpdump, and Nmap
- Linux and infrastructure lab environments (OpenWrt, local servers, cloud Ubuntu)
- Technical documentation and lab write-ups that explain security design decisions
- Self-driven continuous learning across network security, GRC, and OT security foundations

TECHNICAL SKILLS

Security Frameworks & Methods: NIST Cybersecurity Framework (CSF) 2.0, MITRE ATT&CK, risk assessment, threat scenario mapping, incident response planning, security recommendations, quantitative risk / ALE concepts

Network Security: OSI Model, TCP/IP, DHCP, ARP, ICMP, routing, OSPF, VLANs, VLSM, ACLs, NAT/PAT, network segmentation, VPN concepts, TLS, SSH, network design and troubleshooting

Tools & Platforms: Wireshark, Tcpdump, Nmap, Kali Linux, OPNsense, pfSense, Cisco Packet Tracer, Splunk, Wazuh, Keycloak, OpenWrt, Git, GitHub

Systems & Infrastructure: Linux (Ubuntu/OpenWrt), SSH, Bash scripting, PowerShell, Windows, Active Directory fundamentals, UAC, Windows Registry, Windows Defender, BitLocker, local servers, Banana Pi, Raspberry Pi

Programming & Data: Python, Pandas, NumPy, data cleaning, technical documentation

Developing / In Progress: IAM (Keycloak), SIEM deployment and analysis, cloud security, AI/ML for network monitoring, GRC documentation, Docker (foundational exposure)

RELEVANT COURSEWORK

Advanced Network Security: VPNs and cryptography, network troubleshooting, network design, SIEM deployment, MITRE ATT&CK Framework

Integrated Infrastructure Cybersecurity: NIST CSF 2.0 gap analysis, GOVERN function, C-SCRM, incident response playbooks, regulatory reporting chains, Purdue Model, Industrial DMZ, EDR/XDR, Sigma Rules, ZTNA, quantitative risk / ALE, threat surface mapping

SELECTED CYBERSECURITY & INFRASTRUCTURE PROJECTS

Cybersecurity Risk Assessment Project

- Used NIST CSF to assess cybersecurity gaps across governance, protection, detection, response, and recovery areas.
- Applied MITRE ATT&CK to map threat scenarios including social engineering, supply-chain compromise, insider risk, and APT activity.
- Analyzed incidents including Stuxnet, SolarWinds, and Uber to identify lessons for improving cybersecurity resilience.
- Recommended controls including MFA, network segmentation, vulnerability management, incident response planning, supply-chain risk management, and continuous monitoring.

Enterprise Network Security, Firewall & Traffic Analysis Labs | *Advanced Network Security*

- Designed and configured small-scale enterprise networks using VLANs, VLSM subnetting, DHCP, OSPF/static routing, NAT, and ACLs.

- Built segmented LAN, WAN, and DMZ environments and deployed OPNsense firewall rules to control Internal, External, and DMZ traffic flows.
- Performed Nmap comparisons before and after firewall restrictions to demonstrate reduced reconnaissance visibility and minimized exposed attack surface.
- Captured and analyzed network traffic using Wireshark and Tcpdump to inspect protocols, validate connectivity, and support troubleshooting.

Offline Network-Attached Storage (NAS) Architecture | Infrastructure Project

- Engineered a portable, internet-independent NAS using Banana Pi BPI-R4, NVMe storage, OpenWrt, and Wi-Fi 7 for local multimedia file sharing.
- Configured firewall zones and an independent Wi-Fi access point to isolate the local network from external internet dependency.
- Implemented Smart Queue Management (SQM) to reduce bufferbloat and maintain low-latency performance during simultaneous multi-user loads.
- Developed chunked upload functionality using uHTTPd and PHP 8 CGI to support reliable multi-gigabyte transfers without memory exhaustion.

Identity and Access Management Lab | Keycloak

- Configured users, roles, and access control concepts in Keycloak to understand IAM workflows and role-based access foundations.

ML-Based Network Monitoring IDS | Ongoing

- Built and deployed monitoring agents on cloud Ubuntu hosts with a server-hosted dashboard for network and host activity analysis.
- Uses live network captures for prototype monitoring; planning NSL-KDD or newer intrusion detection datasets for ML model optimization.
- Applied Isolation Forest for anomaly detection; remaining work includes full installer deployment, host agent rollout, and ML tuning.
- Demonstrates applied machine learning for detecting possible attacks based on network and host activity.
- GitHub: github.com/benboakye/ML-Network-Intrusion-detection

PROFESSIONAL EXPERIENCE

Intern - Avionics & Systems Diagnostics | Ghana Air Force | Sep 2024 - Oct 2024

- Assisted senior Aircraft Maintenance Engineers during pre-flight checks, engine warming, testing procedures, and inspection of avionics and electronic systems.
- Supported 3 pre-flight checks and performed early morning inspections 3 times per week in a regulated operational environment requiring procedural discipline and safety awareness.
- Accurately tracked and logged departure and arrival data for domestic flight operations, supporting operational accountability and record keeping.
- Applied systems thinking through testing for system resilience and regular maintenance procedures to help ensure system safety.

Data Entry Intern | PRIME Infrastructure & Engineering Services Ltd | Oct 2023 - Dec 2023

- Audited, cleaned, and validated engineering infrastructure data including cell tower material deployment records, inventory, expenditure records, and operational equipment logs.
- Processed approximately 8-12 business operations records per day and 60-80 telecom equipment/material records per week within a 6-person team.
- Supported RF Engineers and Project Managers during site visits for distributing telecom equipment and materials, helping improve distribution speed by about 10%.
- Strengthened data integrity and documentation accuracy for infrastructure-related engineering operations.

EDUCATION

MSc Computer Science - Cybersecurity Operations & Management: Humber Polytechnic | Expected Aug 2027

Bachelor of Science, Telecommunications Engineering: Ghana Communication Technology University | Nov 2025

Telecommunications foundation: embedded systems, communication systems, source-to-channel-to-receiver models, computer systems in data communication, and analog communication systems - supporting network security and OT/infrastructure understanding at the physical and systems level.

CERTIFICATIONS

ISC2 Certified in Cybersecurity (CC): Certified